

Candidate's Name _____

--	--



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2011
Higher 2

H2 BIOLOGY

9648/02

Paper 2 Core Paper

13 September 2011
2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough workings.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination,

1. Fasten all your work securely together.
2. Hand in the following separately:
 - Section A (Part I)
 - Section A (Part II)
 - Section B

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's Use	
Section A	
1	/ 14
2	/ 11
3	/ 12
4	/ 14
5	/ 9
6	/ 10
7	/ 10
Section B	/ 20
Total	/ 100

Section A (Part I)
Answer **all** the questions in this section.

For
Examiner's
Use

QUESTION 1

Fig. 1.1 below depicts an eukaryotic cell.

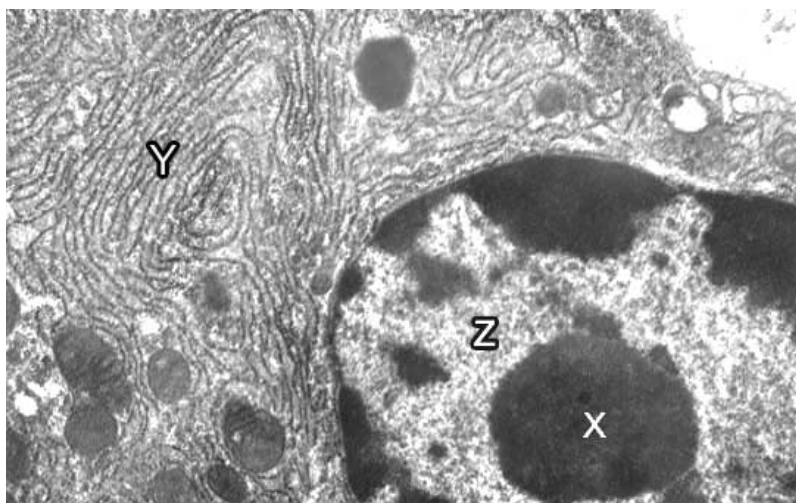


Fig. 1.1

(a) Describe the function of the region labelled 'X'. [1]

.....

.....

(b) Describe two differences between the products of transcription from regions X and Z. [2]

1.
2.

(c) Outline three ways that protein synthesis can be controlled at region Y. [3]

1.
2.
3.

Protein synthesis in a prokaryote is illustrated in Fig. 1.2.

For
Examiner's
Use

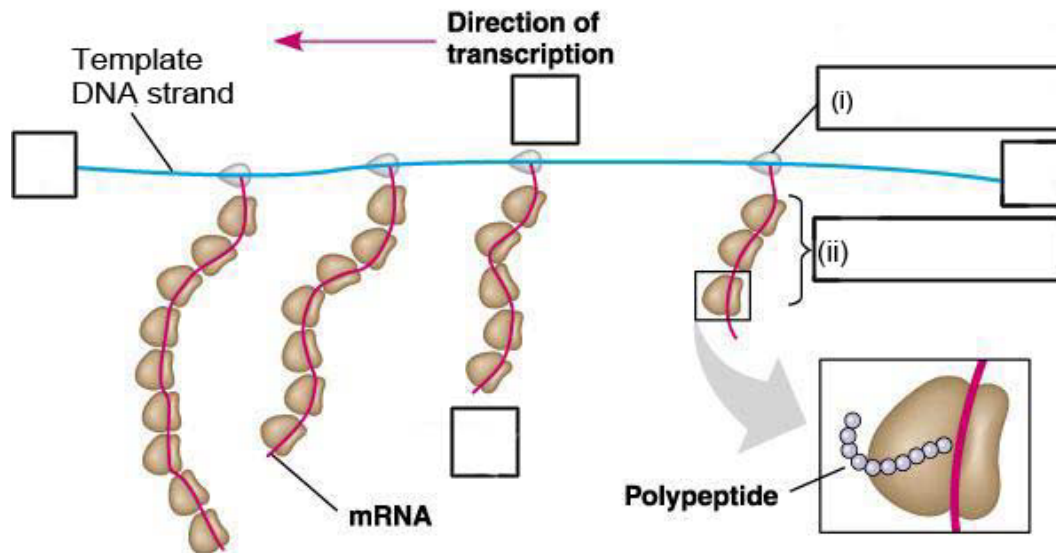


Fig. 1.2

(d) Label the 5' and 3' ends of the mRNA and DNA template strand and structures (i) and (ii) on Figure 1.2. [4]

(e) State the main difference between the processes shown in Fig. 1.2 and that occurring in eukaryotes. [1]

.....

.....

(f) Explain your answers in (e). [3]

.....

.....

.....

.....

.....

[Total: 14]

QUESTION 2

Phosphofructokinase (PFK) is a tetrameric enzyme that plays a central role in controlling the rate of glycolysis. In many organisms, the activity of PFK is enhanced allosterically by several substances, including ADP, and is inhibited allosterically by several other substances, including ATP and citrate. Fig. 2.1 shows part of the glycolysis reactions.

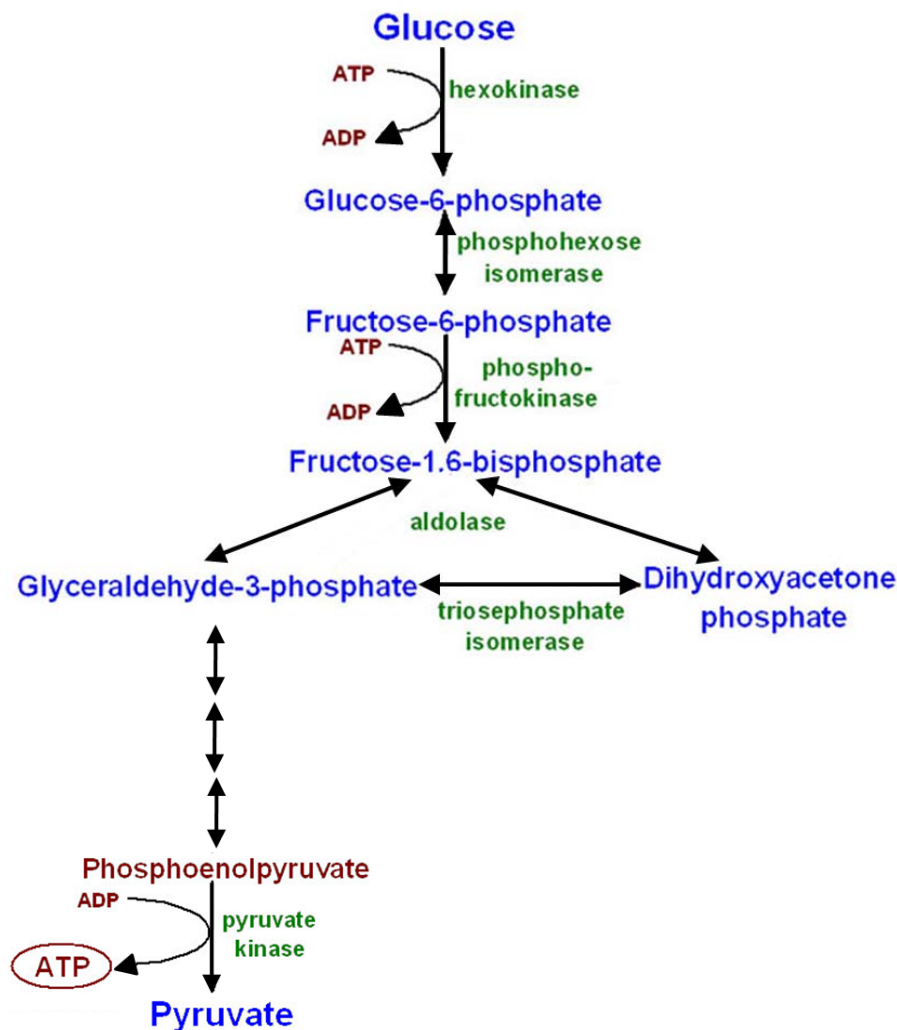


Fig. 2.1

- (a)** Explain why phosphofructokinase is a quaternary protein. [1]

.....

.....

- (b)** Besides PFK, hexokinase and pyruvate kinase also play a role in determining the overall rate of glycolysis.

With reference to Fig. 2.1, suggest why reactions catalyzed by these three enzymes determine the rate of glycolysis. [1]

.....

.....

(c) Explain the importance of inhibition of PFK by citrate when citrate concentration in the cell is high. [2]

.....

.....

.....

(d) Describe how citrate inhibits PFK. [3]

.....

.....

.....

.....

.....

(e) In mammals, under anaerobic condition, both Krebs cycle and oxidative phosphorylation do not occur. Instead, lactate fermentation occurs.

Explain the significance of lactate fermentation. [2]

.....

.....

.....

(f) Hexokinase, the enzyme involved in the first step of glycolysis, is able to utilize various hexoses such as glucose, fructose and mannose.

Suggest how hexokinase is able to utilize different substrates. [2]

.....

.....

.....

[Total: 11]

QUESTION 3

HIV is a retrovirus known to cause acquired immune deficiency syndrome (AIDS).

For
Examiner's
Use

(a) Outline the process by which HIV enters the host cell.

[3]

.....

.....

.....

.....

.....

Long terminal repeats (LTRs) are sequences of DNA that are repeated hundreds or thousands of times. They are found in retroviral genetic material, flanking functional genes. In viruses like HIV, LTRs are necessary for the insertion of viral genetic sequences into host genomes using the enzyme integrase. HIV-1 integrase binds to specific sites at the ends of the viral DNA.

Secondly, LTRs act as promoters that can control transcription when integrated into the host genome. Protein-binding sites in the LTR are involved with transcription. Fig. 3.1 shows a basic model of a retroviral genome.

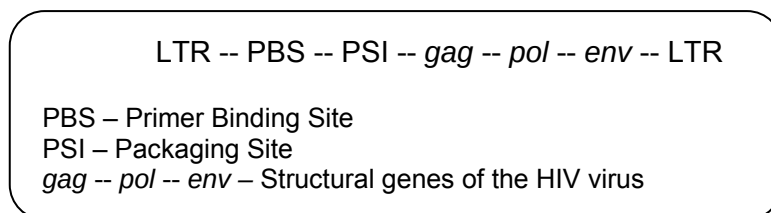


Fig. 3.1

(b) Explain how LTR sequences might function as a promoter for transcription.

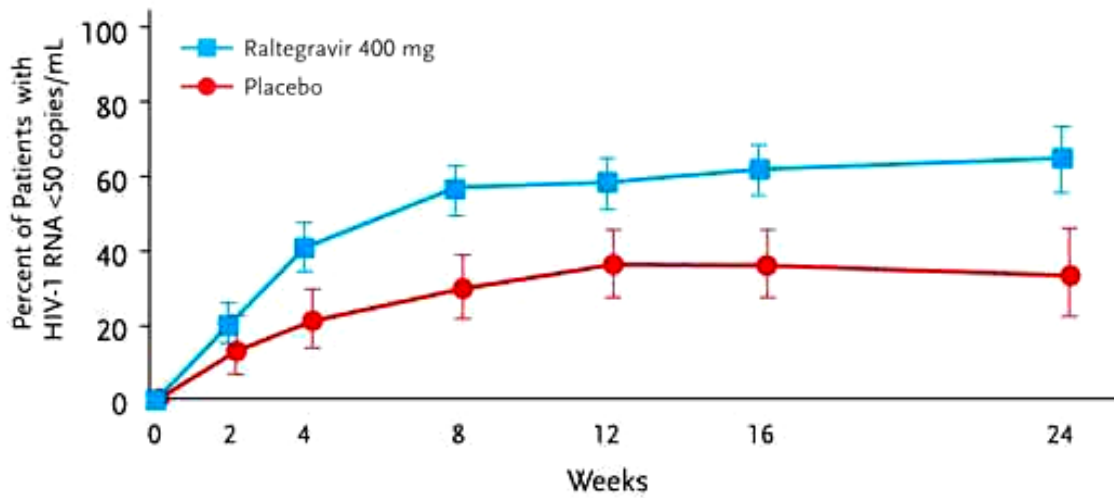
[2]

.....

.....

.....

Many integrase inhibitors have been discovered during the past 10 years, with some of them presently in clinical trials. An example of one such inhibitor is Raltegravir. In one such trial, one group of patients was given Raltegravir while the control group was given placebos (drug-free pills) over 24 weeks. The number of copies of HIV-1 RNA per ml of blood plasma was measured and the results are shown in Fig. 3.2.



Adapted from Merck & Co., Inc. Results of BENCHMRK-1 and-2, two phase-III studies evaluating the efficacy and safety of raltegravir, a novel HIV-1 integrase inhibitor, in patients with triple-class resistant virus. 14th Conference on Retroviruses and Opportunistic Infections; February 25–28, 2007; Los Angeles, CA.

Fig. 3.2

(c) With reference to Fig. 3.2, describe the effect of treating patients with 400mg of Raltegravir. [2]

.....

.....

.....

.....

(d) Suggest how Raltegravir might work as an antiretroviral drug. [2]

.....

.....

.....

Another result of HIV infection is cancer. Individuals infected with HIV have increased risk of three types of tumours: Kaposi's sarcoma, carcinomas of mouth and rectum, and B cell lymphomas disorders.

For
Examiner's
Use

(e) Explain how a virus like HIV may cause cancer. [3]

.....

.....

.....

.....

.....

[Total: 12]

QUESTION 4

Flower color in *Dahlia* flowers is determined by two gene loci, **A/a** and **B/b**.

For
Examiner's
Use

The alleles give the following phenotypes:

A yellow	a white
B white	b yellow

A cross between two white-flowered plants produces offspring of the following phenotypes:

Yellow-flowered plants	67
White-flowered plants	299

(a) Construct a genetic diagram to show the cross described.

[5]

(b) Explain how interaction between the two genes gives rise to white-flowered plants. [4]

.....

.....

.....

.....

.....

.....

(c) Explain the purpose of performing a chi-squared test on the results of the cross. No calculation is required. [2]

.....

.....

.....

.....

(d) Explain how you could determine that a yellow-flowered plant selected from the above cross is pure-breeding. [3]

.....

.....

.....

.....

.....

[Total: 14]

Section A (Part II)
Answer **all** the questions in this section.

QUESTION 5

Bacterial conjugation was first discovered in 1945 when Joshua Lederberg began investigating mating in bacteria. He conducted the following experiment as in Table 5.1 and Fig. 5.1.

Strain A, B and a mixture of both strains were grown, washed in minimal medium (MM) and grown on culture plates containing minimal medium that does not contain essential amino acids. The results are shown in Fig. 5.1.

Mutant genes (–) do not code for enzymes that synthesize amino acids. Note that all five amino acids are required for bacterial growth.

Bacterial strains	Genes for biosynthesis of amino acids	Mutant genes for biosynthesis of amino acids
A	thr ⁺ leu ⁺ thi ⁺	met [–] bio [–]
B	met ⁺ bio ⁺	thr [–] leu [–] thi [–]
Mixtures of Strains A and B	As found in strains A and B	As found in strains A and B

Table 5.1

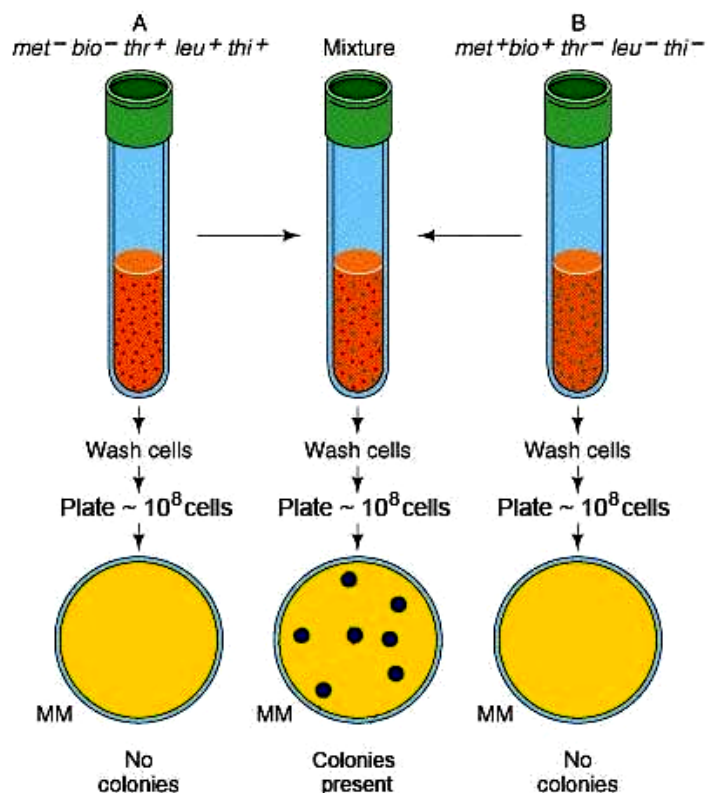


Fig. 5.1

(a) From the experiment in Fig. 5.1, explain the results obtained for

(i) strain A

[2]

.....

.....

.....

.....

(ii) mixture of strain A and B

[3]

.....

.....

.....

.....

.....

(b) Name one other factor that must be present in either Strain A or B for these results to be possible. [1]

.....

(c) Lederberg and his collaborators later discovered viral transduction in 1951.

State three differences between transduction and conjugation.

[3]

1.
2.
3.

[Total: 9]

QUESTION 6

Fig. 6.1 shows the light-dependent stage of photosynthesis.

For
Examiner's
Use

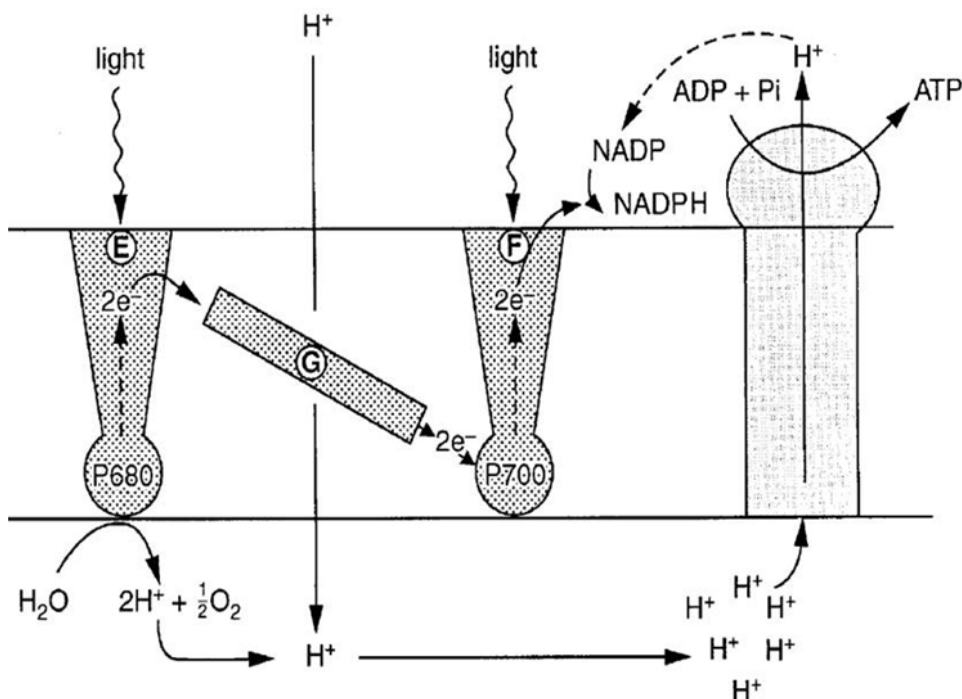


Fig. 6.1

(a) Diquat and paraquat are herbicides (weed killers) that readily accept electrons. They are known to act on the thylakoid membranes.

(i) With reference to Fig. 6.1, explain how these herbicides cause the plants to die. [5]

.....

.....

.....

.....

.....

.....

.....

(ii) Some weed species are not killed when these herbicides are applied to them.

Suggest why this is so.

[1]

.....

.....

(iii) Describe one distinct structural difference between a phospholipid molecule found in the thylakoid membrane and a fat molecule. [1]

For
Examiner's
Use

.....

.....

(b) Plants that live in the Arctic have a relatively short growing season in which the light intensity is always relatively low. Many species growing in these conditions have a high level of anthocyanin, an accessory pigment, in their leaves. The combined effect of these red pigments with the green chlorophylls makes the leaves appear dark purple or black.

Suggest why this adaptation is useful in increasing photosynthetic rates. [3]

.....

.....

.....

.....

.....

[Total: 10]

QUESTION 7

Mitochondrial DNA analysis of various rodents was done and their phylogenetic relationships represented by Fig. 7.1

For
Examiner's
Use

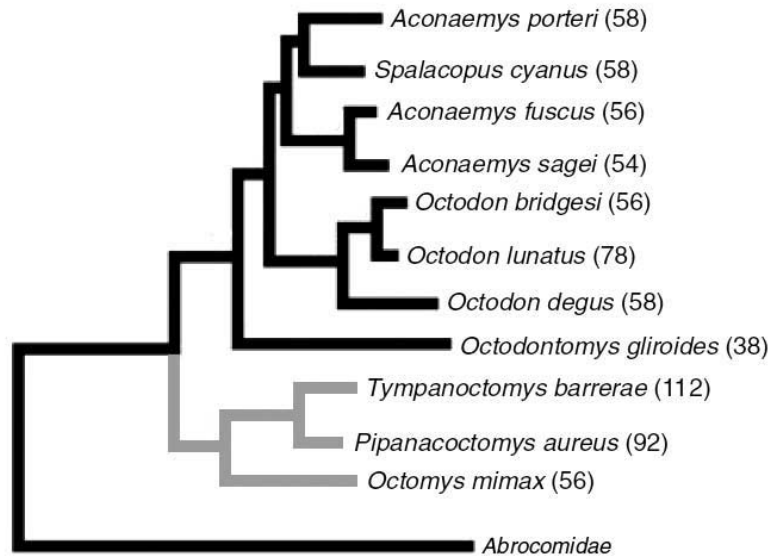


Fig. 7.1

- (a) Suggest why mitochondrial DNA was used to study the phylogenetic relationships between the various species of rodents. [2]

.....

.....

.....

- (b) Explain how DNA analysis of mitochondrial DNA can provide evidence for the relationships shown in Figure 7.1. [2]

.....

.....

.....

Cases of mammalian tetraploidy are rare. The Plains Viscacha Rat, *Tympanoctomys barrerae*, is one of the few polyploid mammals in existence and it is tetraploid ($4n$) with 112 chromosomes. It is only found in the desert scrublands of the central western plains in Argentina. It was hypothesised that *Tympanoctomys barrerae* evolved from the same common ancestor as *Octomys mimax*, a diploid rodent with 56 chromosomes ($2n$). *O. mimax* inhabits rocky desert and semi-desert environments up to 800 metres above sea level.

(c) Explain how tetraploidy may have first developed in the ancestral rat population. [2]

.....

.....

.....

.....

(d) Describe how *Tympanoctomys barrerae* may have evolved as a separate species as a result of natural selection. [4]

.....

.....

.....

.....

.....

.....

.....

[Total: 10]

Section B

Answer **one** question.

For
Examiner's
Use

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in questions (a), (b), etc., as indicated in the question.

QUESTION 8

- (a) Describe the molecular structure of cellulose found in the cell wall of a plant. [6]
- (b) Explain how ATP enables nerve cells to transmit nerve impulses. [6]
- (c) Explain the homeostatic mechanism in response to a rise in blood glucose concentration. [8]

QUESTION 9

- (a) Describe the main structural features of the *lac* operon. [8]
- (b) Explain how mitosis maintains genetic stability. [6]
- (c) Distinguish between the processes of meiosis I and meiosis II in eukaryotes. [6]

[Total: 20]

● End of Paper 2 ●